INSURANCE CHARGES DASHBOARD

in the United States

smoker

region

Total Customers

1,338

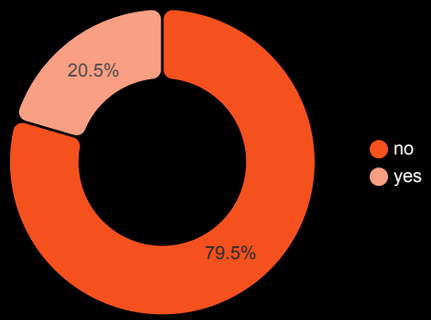
Average Charges

13.3K

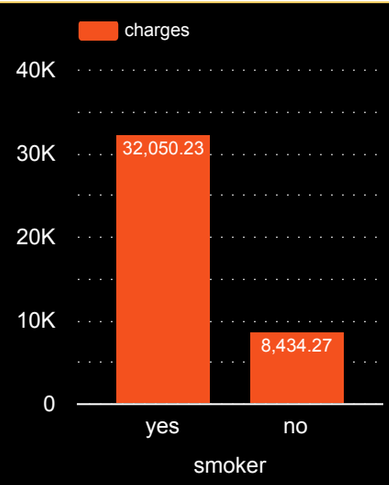
Average Age

39.21

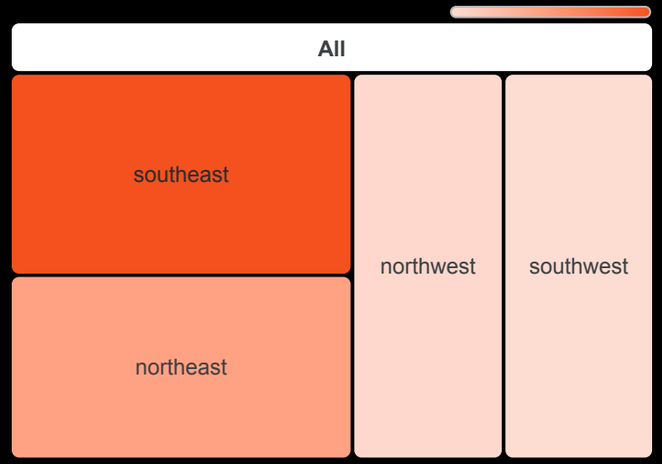
smoker



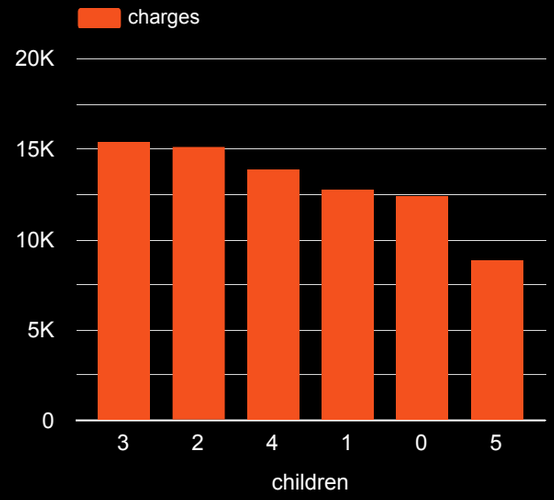
charges by smoker



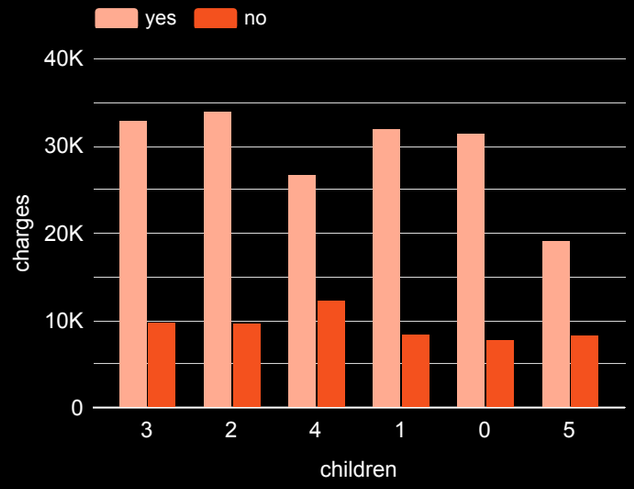
charges by region



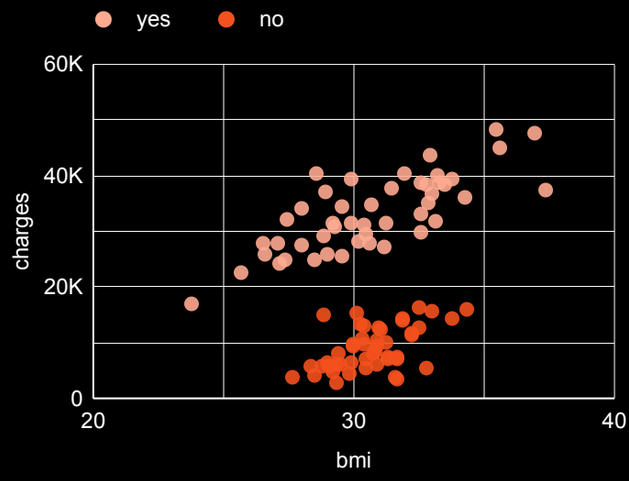
charges by children



charges by children and smoker



BMI vs Charges by Smoking Status



Prediction Results of the MLR Model for Insurance Charges in the United State.

OLS Regression Results

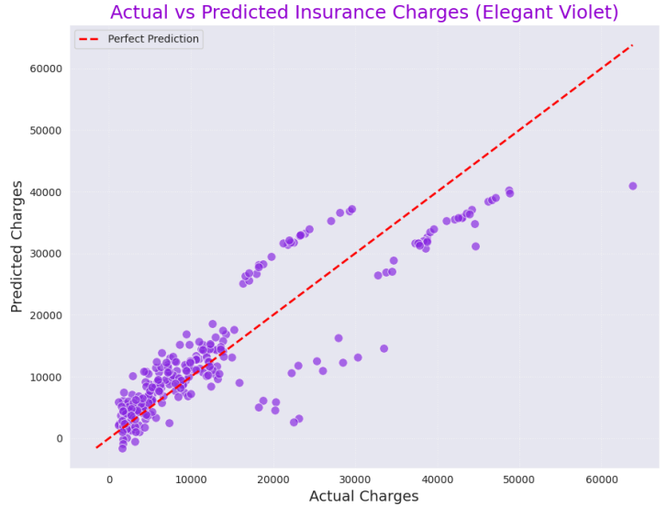
Dep. Variable:	charges	R-squared:	0.742
Model:	OLS	Adj. R-squared:	0.740
Method:	Least Squares	F-statistic:	380.9
Date:	Thu, 12 Mar 2026	Prob (F-statistic):	1.32e-305
Time:	09:09:09	Log-Likelihood:	-10845.
No. Observations:	1070	AIC:	2.171e+04
Df Residuals:	1061	BIC:	2.175e+04
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-1.193e+04	1114.505	-10.705	0.000	-1.41e+04	-9744.335
age	256.9757	13.477	19.067	0.000	230.530	283.421
bmi	337.0926	32.471	10.381	0.000	273.378	400.807
children	425.2788	154.655	2.750	0.006	121.814	728.743
sex_male	-18.5917	376.175	-0.049	0.961	-756.722	719.539
smoker_yes	2.365e+04	466.505	50.699	0.000	2.27e+04	2.46e+04
region_northwest	-370.6773	536.873	-0.690	0.490	-1424.130	682.776
region_southeast	-657.8643	539.791	-1.219	0.223	-1717.043	401.314
region_southwest	-809.7994	535.208	-1.513	0.131	-1859.986	240.387

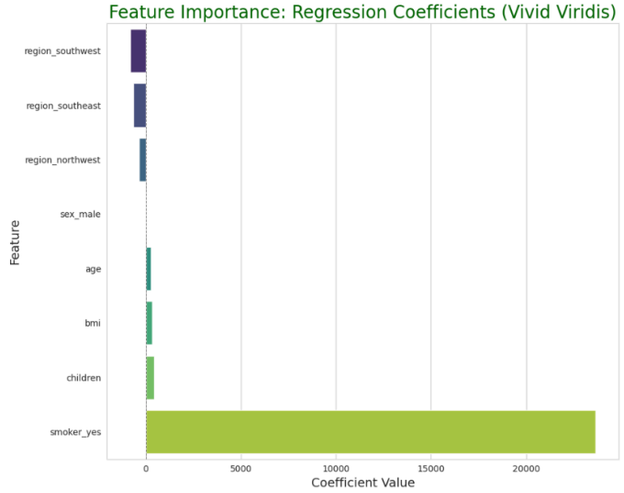
Omnibus:	252.330	Durbin-Watson:	2.085
Prob(Omnibus):	0.000	Jarque-Bera (JB):	613.798
Skew:	1.253	Prob(JB):	5.19e-134
Kurtosis:	5.737	Cond. No.	310.

------ MODEL PERFORMANCE ------

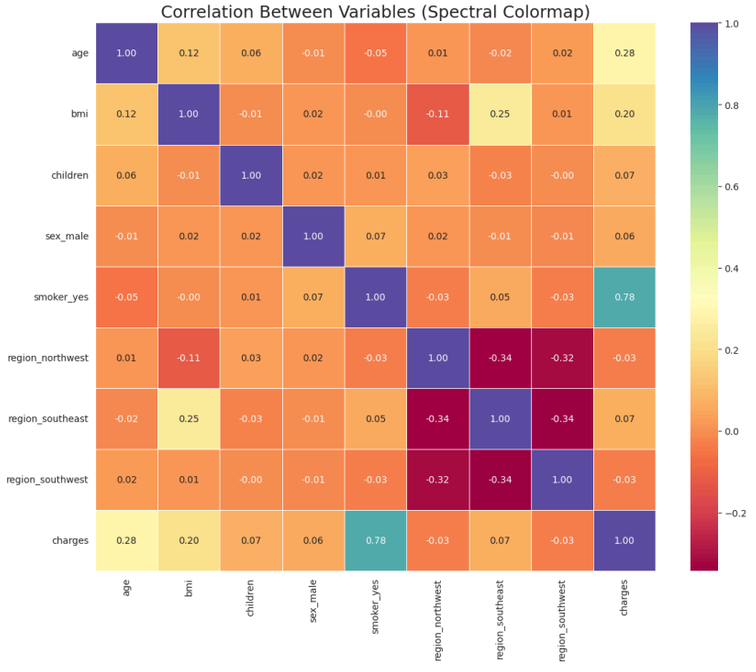
	Metric	Train	Test
0	RMSE	6105.545160	5796.284659
1	MAE	4208.234572	4181.194474
2	MAPE	42.202685	46.888256
3	R2	0.741726	0.783593



Summary of Model Analysis and Actual vs Predicted Graph
- The regression model explains about 74% of the variance in insurance charges, indicating a reasonably good predictive performance.
- From the Actual vs Predicted graph, the model predicts insurance costs fairly well for low to medium charge values, but the prediction error increases for higher charges.

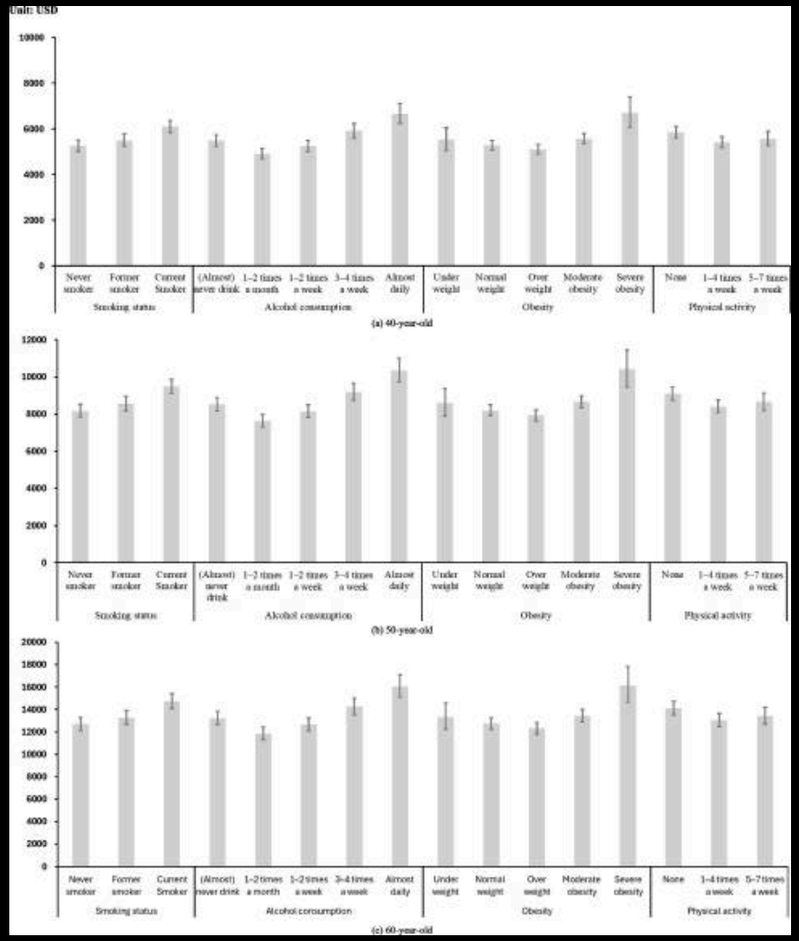


Feature Importance (Regression Coefficients) – Summary
- Smoking (smoker_yes) has the strongest impact on insurance charges.
- Children, BMI, and age increase charges but with much smaller effects.
- Male gender and some regions show negative coefficients, indicating slightly lower predicted costs.

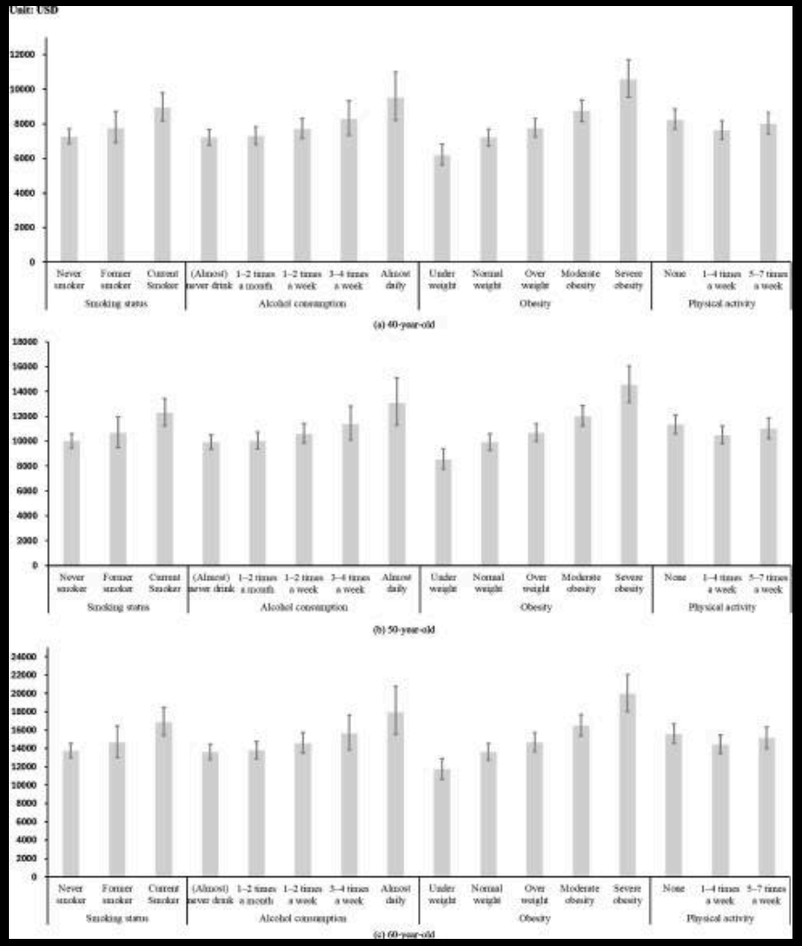


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The effects of smoking, alcohol consumption, obesity, and physical inactivity on healthcare costs: a longitudinal cohort study



Cumulative healthcare costs for **males** by health risk factors at ages 40, 50, and 60



Cumulative healthcare costs for **females** by health risk factors at ages 40, 50, and 60





Study population : 62,965
Men : 35,107
Women : 27,858
Follow-up from : 2010 - 2019

alternative data โดยสรุป งานวิจัยนี้สนับสนุนผลการศึกษาของโมเดล regression ที่พบว่า ปัจจัยด้านพฤติกรรมสุขภาพ โดยเฉพาะการสูบบุหรี่และภาวะโรคอ้วน มีผลอย่างมีนัยสำคัญต่อค่าใช้จ่ายด้านสุขภาพ การค้นพบดังกล่าวแสดงให้เห็นว่าการปรับเปลี่ยนพฤติกรรมสุขภาพ เช่น การเลิกสูบบุหรี่ การควบคุมน้ำหนัก และการออกกำลังกาย สามารถช่วยลดภาระค่าใช้จ่ายด้านสุขภาพในระยะยาวได้